AISIN Europe Group • College of Computing

Smart Urban Waste Management

Using Truck-Mounted Cameras to Clean Cities Better

March 2026 • Phase 1 Review & Phase 2 Roadmap

Why This Problem Is Of Interest

We live in Morocco. We encounter this problem every day.

An overflowing bin is not an exceptional event in our neighbourhoods — it is background noise. Waste on the pavement, bags torn open overnight, the smell of organic matter on a hot afternoon. We have learned, almost without noticing, to walk past it.

That normalisation is the most insidious part.

When a bin overflows every week without consequence: residents stop reporting, the municipality stops hearing, and the cleaning company faces no pressure to change.

We know which corners of our cities are reliably dirty. Which market days bury certain streets in waste. Which specific bins overflow every single week. This is granular, lived knowledge — invisible to any official system. One of the things we are building is exactly the infrastructure to capture it.

The street stays dirty — not through neglect exactly, but through the **absence of any feedback loop.**

What It Actually Costs

A health hazard at ground level

In dense residential areas, the street is where children play. Broken glass, sharp metal lids, and rotting matter around an overflowing bin become part of that space — a recurring condition in many Moroccan cities.

Overflow that compounds itself

Stray animals tear into bags left on the ground. What begins as a localised overflow scatters across a wide radius overnight. By the time a truck arrives, it is a manual cleanup operation — slower, more expensive, entirely avoidable.

Drainage failure and flooding

Waste accumulating near gutters blocks water flow, particularly after rain. In older neighbourhoods with limited drainage infrastructure, this directly contributes to street flooding — a secondary hazard born from an unresolved primary one.

An unequal burden by district

The problem does not fall evenly. Denser, lower-income areas — already underserved in bin density and collection frequency — bear the most visible impact. The current system has no mechanism to detect, let alone correct, this gap.

The Big Picture — What We Have & What We Want

TODAY	WITH OUR SYSTEM
Fixed schedule, designed once	Multi-fleet cameras observe continuously
Trucks visit all bins equally	System prioritises by actual need
No feedback — overflows unseen	Overflows detected and flagged early
Municipality relies on complaints	Municipality tracks performance by data

this project

The System — Four Connected Layers

Layer 1 — See

Camera captures bin states, street conditions, displaced bins

Layer 2 — Understand

Visual analysis classifies each observation, geo-tags each event

Layer 3 — Respond

Routing engine reprioritises trucks based on real-time demand

Layer 4 — Oversee

Municipality dashboard tracks KPIs, incidents, and service quality

Three problems, one system

- **We don't know what the streets look like** — no live city-wide picture of bin states between collections
- **Truck routes are blind to actual need** — fixed schedules visit empty bins while others overflow
- **The municipality has no objective feedback** — oversight relies on complaints and self-reporting

Problem 1

We Don't Know What the Streets Look Like

Problem 1 — The Visibility Gap

How cities currently find out about problems:

- Citizen complaint — slow, patchy, skewed toward wealthier areas
- Manual inspection — expensive and too infrequent to matter
- Truck driver noticing in passing — informal and undocumented

> All three are reactive. None produce structured, reusable data.

What the camera changes

Dashcam video stream

↓

Object detection per frame

↓

Classify & geo-tag each event

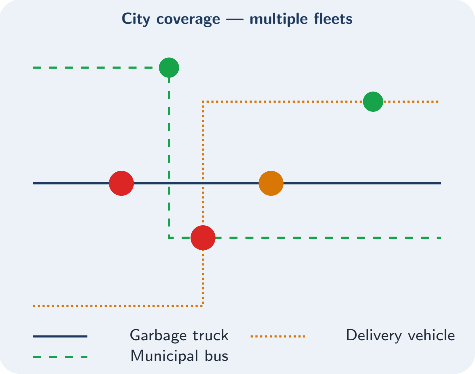
↓

Live bin log — timestamped & located

Extending Coverage — Multiple Fleets

The same camera-and-GPS setup can be extended to other vehicle fleets:

- **Municipal buses** — follow fixed routes multiple times per day, covering main arteries far more frequently than any dedicated cleaning fleet
- **Delivery vehicles** — penetrate residential side streets and market areas on a daily basis
- **Street-sweeping machines** — already operate in dense zones; lowest-friction fleet to instrument given their direct link to waste management



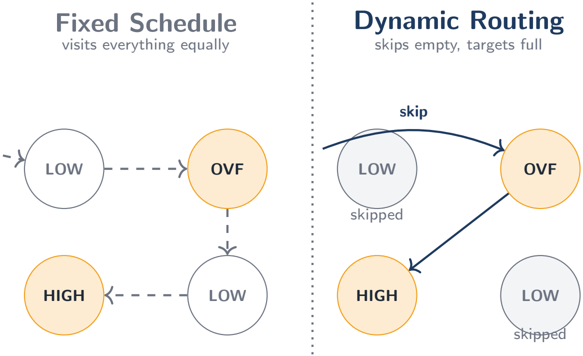
Problem 2

Truck Routes Are Blind to Actual Need

Problem 2 — Blind Routing

A quiet residential bin is collected twice a week averaging LOW fill. A bin outside the central market reaches OVERFLOW every Tuesday afternoon. Both are on identical schedules. The system is deployed where it is least needed.

Combining camera observations with fleet GPS gives us demand data: which bins fill fastest, which zones are consistently high-load, and where trucks cover ground without finding much to collect.



How We Measure Success

KPIs Across Three Categories

KPIs — Three Categories, One System

Cleanliness	Efficiency	Accountability
Overflow Rate	Kilometres per Tonne Collected	Coverage Rate
Fill Class at Collection	Unnecessary Collection Rate	Citizen Complaint Volume
Time-to-Resolution	Fleet Utilisation	
Street Cleanliness Score		

> If overflow rate drops 40% but kilometres per tonne *increases*, the system is preventing overflows by dispatching more trucks — not by routing smarter.

Phase 2 — What We Are Building Next

Data Pipeline, Annotation & Detection MVP

The Two Steps We Are Covering Now

Step 1

Data Pipeline & Annotation

→

Step 2

Detection MVP

These two steps are the core deliverable of the next 1–2 weeks.

Everything in Phase 3 (routing optimisation, live dashboard) depends on getting them right first.

The Five Fill-Level Classes

Class	Operational meaning	Action
EMPTY	Bin visibly empty or near-empty; collection unnecessary	No action — skip
LOW	Up to approximately one-third full	Skip
MEDIUM	One-third to two-thirds full	Schedule normally
HIGH	Two-thirds to rim; approaching capacity	Collect soon
OVERFLOW	Waste visible above rim or on the ground around the bin	Dispatch now

Annotation Guidelines & Process

Before any labelling begins, we define:

1. **Reference sheet** — at least 5 representative images per class, covering varied lighting, bin types, camera angles, and partial occlusion
2. **Edge-case rules** — explicit decisions for ambiguous frames:

   Open lid with partial fill → HIGH

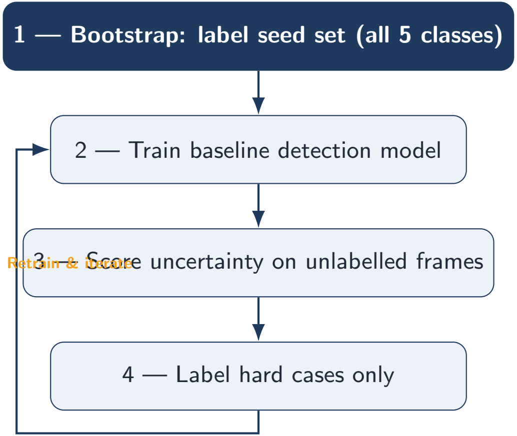
   Bin obscured by parked vehicle → unobservable, excluded from training

Tooling candidates

- **Label Studio** — open source, bounding box + classification, exports to COCO / YOLO
- **CVAT** — open source, strong multi-annotator workflow, widely used in computer vision research

Final choice depends on the infrastructure preferences.

Active Learning Strategy



Why this loop?

- Avoids labelling thousands of frames upfront
- Explicitly targets the model's failure modes: night conditions, partial occlusion, unusual bin positions, non-standard bin types
- Each cycle yields a stronger model with fewer total annotations than random labelling would require

Detection MVP — What We Are Building

The MVP does exactly this

Input: video frame from truck camera

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Detect: bounding box around each bin

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Output: fill-level class per detected bin

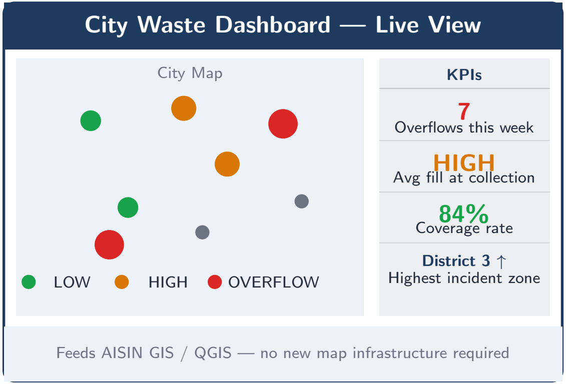
Model candidates

- **RT-DETR** — real-time capable, strong on small objects in street scenes

Evaluation metrics

Precision	Avoid false overflow alerts
Recall	Don't miss actual overflows
Per-class F1	Focus on OVERFLOW & HIGH
mAP	Overall detection quality

A First Look at the Dashboard



Map colour coding

- Green — LOW / EMPTY
- Orange — HIGH, collect soon
- Red — OVERFLOW, dispatch now

Timeline — The Next 1–2 Weeks

Week 1 — Foundation
Define annotation guidelines & reference sheet
Label seed set — all 5 classes, diverse conditions
Inter-annotator agreement check (Cohen's κ)
Train baseline detection model on seed set

Week 2 — MVP
Active learning cycle — flag & label hard cases
Retrain & evaluate on held-out test set
Ground-truth validation on one real route
Failure mode documentation & results summary